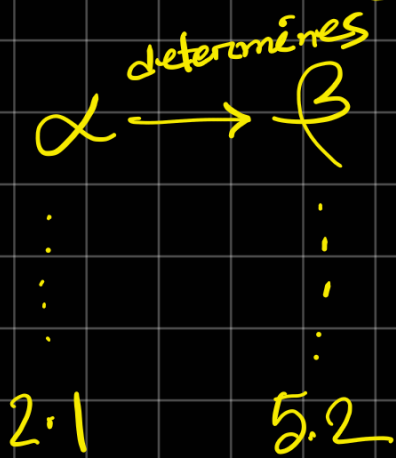


Functional dependency

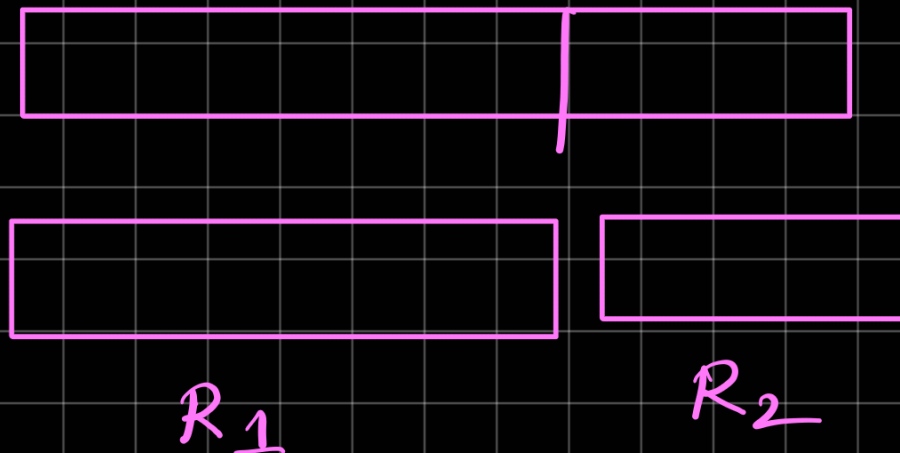


if we get 2.1 guaranteed to get 5.2
not vice versa.

2.3 5.2 ✓

3.1 5.1

2.1 5.2 ✓ R



splitting $\rightarrow$ schema decomposition.

$$F = \begin{bmatrix} \alpha \rightarrow \beta \\ \beta \rightarrow \delta \\ \delta \rightarrow \epsilon \end{bmatrix}$$

$\alpha \rightarrow \delta$ (Rule of Transitivity?)

$$F^+ = \begin{bmatrix} \alpha \rightarrow \beta \\ \beta \rightarrow \delta \\ \delta \rightarrow \epsilon \\ \alpha \rightarrow \delta \end{bmatrix}$$

R	EID	Name	DOB	P N	PTD	PED

↑ issue

↓ Expire

TCP/IP

→ 2s, 4s, 8s double up
waiting time.
Most efficient.

NID

CID	cName	DOB	PN
-----	-------	-----	----

↓
FK

Passport

PN	PID	PEN
----	-----	-----

$$R_1 \cap R_2 = PN$$

PN → passport

NID

CID	cName	DOB
-----	-------	-----

Passport

CID	PN	PID	PEN
-----	----	-----	-----

$$R_1 \cap R_2 = CID$$

CID → (NID)

CID → CID, cName, DOB

101 → 101, a, 1-1-01

102 → 102, b, 1-1-02

lossless Decomp.